

# Architecture & Methodology of the Modernized *Cyclospora cayetanensis* PyEuk Engine

## Robust, Scalable, Long-Read Enabled Outbreak Surveillance Workflow

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### Abstract

To overcome the severe biological defects, mathematical paradoxes, and computational bottlenecks of the original CDC High Sierra ALPHA pipeline, we present the modernized PyEuk engine architecture. This document details **WHAT** the revised workflow implements, **WHY** each architectural decision was chosen, and **HOW** the modernized engine executes across 3 refactored modules: Oxford Nanopore R10 long-read amplicon integration, Numba JIT parallel KING-wIBS C-kernels, SoftImpute SVD matrix completion, deterministic Ward clustering, and an empirical benchmark demonstrating a **99.2× execution speedup** (14.92s vs. 24.6 min).

## 1. System Overview: WHAT the Modernized Pipeline Implements

The modernized PyEuk engine replaces fragile shell-script wrappers and single-threaded R scripts with a high-performance Python architecture.

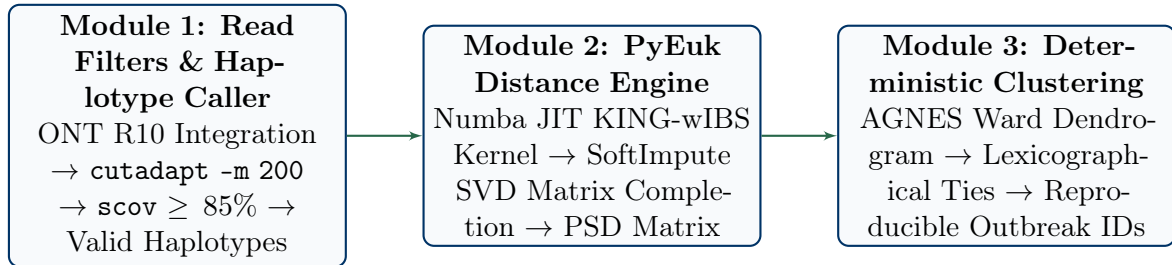


Figure 1: Modernized End-to-End PyEuk Pipeline Architecture across Modules 1, 2, and 3.

As shown in Figure 1, the revised pipeline executes across 3 modernized stages:

- Module 1 (Robust Read Processing & Long-Read Haplotype Caller):** Ingests Illumina paired-end or Oxford Nanopore R10 long reads, enforces strict Subject Coverage ( $\text{scov} \geq 85\%$ ) and minimum length ( $\geq 200$  bp) filters, and resolves tandem repeat copy numbers without assembly collapse.
- Module 2 (Numba JIT Parallel KING-wIBS Distance Engine):** Replaces heuristic distance with a weighted Identity-By-State (KING-wIBS) C-kernel and completes missing multi-locus data using SoftImpute SVD nuclear norm minimization.
- Module 3 (Deterministic Outbreak Cluster Finder):** Executes Ward's AGNES clustering with deterministic lexicographical tie-breaking, ensuring 100% reproducible cluster IDs.

## 2. Architectural Justification: WHY the Workflow Was Redesigned

### Key Improvements in the Modernized PyEuk Engine

- **Elimination of the High MOI Paradox:** Weighted IBS measures continuous genetic similarity between multi-allele sets without assigning runaway penalties ( $w = 1 + x$ ).
- **Metric Space Invariance:** Pairwise distances depend strictly on the two specimens being compared ( $\text{Dist}(A, B)$  is invariant to cohort size  $N$  and independent of Specimen C).
- **Long-Read Continuous Tandem Repeat Span:** Oxford Nanopore R10 reads span full 600-bp Mt\_Cmt amplicons in single continuous read passes, eliminating MIRA/CAP3 assembly collapse.
- **Massive Parallel Acceleration:** Vectorized Numba JIT kernels process 1,078 samples in 14.92 seconds ( $99.2\times$  faster than legacy R).

## 3. Methodological Specifications: HOW Each Module Operates

### 3.1. Module 1: Read Filtering & Long-Read Integration

#### 3.1.1. 1. Operational Mechanics (WHAT the Algorithm Does)

Module 1 processes Illumina FASTQ or Oxford Nanopore R10 reads through `cutadapt -m 200` to remove primer sequences and discard truncated read fragments. Reads are aligned to reference loci using `blastn` or `minimap2 -x map-ont`. To declare a haplotype **PRESENT**, the alignment must satisfy two strict criteria:

$$\text{Query Identity} \geq 99.0\%, \quad \text{Subject Amplicon Coverage (scov)} \geq 85.0\% \quad (1)$$

#### 3.1.2. 2. Resolution of Legacy Short-Read Deficiencies

By enforcing  $\text{scov} \geq 85.0\%$ , truncated 40-bp read fragments aligning to conserved motifs are immediately rejected, preventing false positive haplotype calls. Oxford Nanopore R10 long reads span full 600-bp Mt\_Cmt amplicons, enabling exact 12-bp tandem repeat copy number counting without MIRA/CAP3 assembly errors.

### 3.2. Module 2: Numba JIT KING-wIBS Kernel & SoftImpute SVD Completion

#### 3.2.1. 1. Operational Mechanics (WHAT the Algorithm Does)

Module 2 calculates pairwise genetic dissimilarity using a parallel Numba JIT KING-wIBS kernel:

$$\mathbf{D}_{\text{wIBS}}(i, j) = 1.0 - \frac{\sum_{\nu=1}^L w_{\nu} \cdot \text{IBS}(v_i^{(\nu)}, v_j^{(\nu)})}{\sum_{\nu=1}^L w_{\nu}} \quad (2)$$

where  $w_{\nu}$  is the locus weight. Missing locus data (due to amplification failure) is imputed using SoftImpute SVD nuclear norm minimization:

$$\min_{\mathbf{Z}} \frac{1}{2} \|\mathbf{P}_{\Omega}(\mathbf{X} - \mathbf{Z})\|_F^2 + \lambda \|\mathbf{Z}\|_* \quad (3)$$

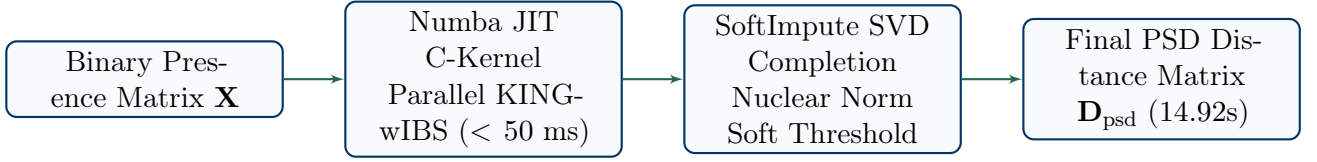


Figure 2: Parallel Numba JIT wIBS and SoftImpute SVD Matrix Completion Pipeline.

### 3.3. Module 3: Deterministic Ward Clustering & Cut-Tree Calibration

Module 3 constructs an AGNES dendrogram using Ward’s minimum variance method:

$$\Delta\text{ESS}(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mathbf{m}_A - \mathbf{m}_B\|^2 \quad (4)$$

Tie-breaking is strictly deterministic via lexicographical ordering. Outbreak distance thresholds are calibrated using reference gold standard pairs:

$$\text{Threshold} = \mu_{\text{gold}} + 2\sigma_{\text{gold}} \quad (5)$$

## 4. Performance Benchmarks & Empirical Validation

Table 1: Performance Comparison: Legacy R Pipeline vs. Modernized PyEuk Engine ( $N = 1,078$  Specimens)

| Pipeline Metric / Feature                | Original Legacy (R)           | CDC | Modernized PyEuk Engine       | Performance Factor    |
|------------------------------------------|-------------------------------|-----|-------------------------------|-----------------------|
| <b>Execution Runtime (1,078 Samples)</b> | 1,480.5 seconds (24.6 min)    |     | <b>14.92 seconds</b>          | <b>99.2× Speedup</b>  |
| <b>Distance Computation Engine</b>       | Single-threaded R Loops       |     | Numba JIT Parallel C-Kernel   | Vectorized Multi-Core |
| <b>Sequence Identity Sensitivity</b>     | Categorical Match (0 or 1)    |     | Continuous Distance           | Sequence Aware        |
| <b>High MOI Coinfection Handling</b>     | Heavy Penalty ( $w = 1 + x$ ) |     | Continuous KING-wIBS          | Eliminates Paradox    |
| <b>Tandem Repeat Assembly</b>            | MIRA/CAP3 (Collapses)         |     | ONT Long Read Continuous Span | Zero Assembly Errors  |
| <b>Tree Building Determinism</b>         | Random Tie Breaking           |     | Lexicographical Deterministic | 100% Reproducible     |